Biomarker–Treatment Correlation During Carryover: A Modeling Artifact in the pmsimstats Data-Generating Process

pmsimstats team

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Abstract

The `pmsimstats` package (Hendrickson et al., 2020) simulates clinical trial data by drawing participant outcomes from a multivariate normal distribution whose correlation structure encodes a biomarker–treatment interaction. A post-publication commit introduced a change to the rule governing how the biomarker–biologic response (BM–BR) correlation is assigned during off-drug periods with carryover residual. This white paper demonstrates that the original correlation rule creates a modeling artifact: it applies full biomarker predictive power to timepoints where the drug effect has decayed to near zero, injecting spurious biomarker-specific variance into off-drug periods. This artifact, not the carryover residual itself, is the primary driver of the power loss reported in the publication. This document traces the mathematics of the artifact, evaluate both the original and revised correlation rules against clinical and statistical criteria, and propose a principled resolution.

Architecture scope. The correlation artifact documented here is specific to Architecture B (MVN differential correlation), in which the biomarker-BR correlation is assigned as a step function of treatment status. Under Architecture A (direct mean moderation), the biomarker-treatment interaction is additive in the mean structure and does not depend on the correlation assignment logic. For a comparison of both DGP architectures, see `02-dgp-mean-moderation-vs-mvn.tex`. For implementation details of Architecture A, see `19-mean-moderation-implementation-notes.tex`.

Contents

1 Introduction **2**

2 The Data-Generating Process **2**
- 2.1 Multivariate Normal Framework . . . 2
- 2.2 Mean Vector Construction . . . 2
- 2.3 Covariance Matrix Construction . . . 3

3 The Two Correlation Rules **3**
- 3.1 Original Rule (Commit `42ac030`) . . . 3
- 3.2 Revised Rule (Commit `8609f12`) . . . 3

4 The Artifact: Conditional Expectations Under the Original Rule **4**
4.1 Conditional Mean of BR Given Biomarker . . . 4
4.2 Numerical Example: N-of-1 Path A at BD3 . . . 4
4.3 The Source of the Artifact . . . 4
4.4 Why This Destroys Power . . . 5

5 Clinical Evaluation **5**
5.1 Is the Original Rule Clinically Reasonable? . . . 5
5.2 Is the Revised Rule Clinically Reasonable? . . . 5
5.3 Summary of Clinical Assessment . . . 6

6 Statistical Evaluation **6**
6.1 Covariance Matrix Coherence . . . 6
6.2 Impact on the Covariance Matrix Positive Definiteness . . . 6
6.3 Effect Size Bias . . . 7
6.4 Type I Error . . . 7

7 Implications for Published Results **7**
7.1 Which Results Are Affected? . . . 7
7.2 Is the Paper's Central Conclusion Still Valid? . . . 7
7.3 Recommendation . . . 7

8 A Principled Correlation Decay Model **8**

9 Conclusions **8**

1 Introduction

The central finding of Hendrickson et al. (2020) is that the N-of-1 trial design, while offering superior power under ideal conditions, is acutely sensitive to carryover effects: even a half-life of 0.1 weeks causes power to drop from 0.95 to 0.18 at $N = 35$. This finding has direct implications for trial design decisions in precision medicine.

A post-publication commit (`8609f12`) by co-author Ron Thomas modified the rule governing how the biomarker–biologic response (BM–BR) correlation is assigned during off-drug periods. Under the revised rule, the power loss from carryover is eliminated entirely. The question of which rule is correct is not a coding error to be fixed but a substantive modeling choice with different implications for the data-generating process (DGP).

This white paper provides a mathematical analysis of why the original rule produces the power drop, whether the drop reflects a genuine statistical phenomenon or a modeling artifact, and what the implications are for the published results.

2 The Data-Generating Process

2.1 Multivariate Normal Framework

Each simulated trial draws N participants from a multivariate normal distribution:

$$\mathbf{X}_i \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

where $\mathbf{X}_i$ is a vector of length $2 + 3K$ (K = number of timepoints), containing the biomarker (BM), baseline symptoms (BL), and three response factors (TV, ER, BR) at each timepoint. The observed outcome is:

$$S_{i,t} = \mathrm{BL}_i - (\mathrm{TV}_{i,t} + \mathrm{ER}_{i,t} + \mathrm{BR}_{i,t})$$

The biomarker–treatment interaction is encoded not in the mean vector $\boldsymbol{\mu}$ but in the covariance matrix Σ. Specifically, the correlation between the biomarker and the BR factor at timepoint t determines the extent to which participants with different biomarker values have different drug responses.

2.2 Mean Vector Construction

The BR mean at timepoint t is computed from the modified Gompertz function evaluated at the cumulative time on drug (tod):

$$\mu^{\mathrm{raw}}_{\mathrm{BR},t} = f_G(\mathrm{tod}_t; \max, \mathrm{disp}, \mathrm{rate})$$

where $f_G(0) = 0$ by construction. When a participant is off drug, $\mathrm{tod}_t = 0$, so $\mu^{\mathrm{raw}}_{\mathrm{BR},t} = 0$.

Carryover is applied sequentially:

$$\mu^{\mathrm{adj}}_{\mathrm{BR},t} = \mu^{\mathrm{raw}}_{\mathrm{BR},t} + \mu^{\mathrm{adj}}_{\mathrm{BR},t-1} \cdot \left(\tfrac{1}{2}\right)^{s \cdot t_{sd}/t_{1/2}}$$

where t_{sd} is time since discontinuation, $t_{1/2}$ is the carryover half-life, and s is a scale factor (1 in the original code, 2 after the Ron Thomas edit).

2.3 Covariance Matrix Construction

The covariance matrix is built from a correlation matrix R and a vector of standard deviations $\boldsymbol{\sigma}$:

$$\Sigma = \mathrm{diag}(\boldsymbol{\sigma}) \cdot R \cdot \mathrm{diag}(\boldsymbol{\sigma})$$

The correlation between BM and BR_t is the critical element. The two code versions assign this correlation differently, as described in the following sections.

3 The Two Correlation Rules

3.1 Original Rule (Commit `42ac030`)

The original code assigns the BM–BR correlation using a simple threshold test on the adjusted BR mean:

$$\rho(\mathrm{BM}, \mathrm{BR}_t) = \begin{cases} c_{bm} & \text{if } \mu^{\mathrm{adj}}_{\mathrm{BR},t} \neq 0 \\ 0 & \text{if } \mu^{\mathrm{adj}}_{\mathrm{BR},t} = 0 \end{cases}$$

This is a step function: any nonzero adjusted BR mean, no matter how small, triggers the full correlation c_{bm}.

In the R code:

```
if (means[which(n1 == labels)] != 0) {
  correlations[n1, 'bm'] <- modelparam$c.bm
  correlations['bm', n1] <- modelparam$c.bm
}
```

3.2 Revised Rule (Commit `8609f12`)

The Ron Thomas revision introduces a three-way assignment with proportional scaling during carryover periods:

$$\rho(\mathrm{BM}, \mathrm{BR}_t) = \begin{cases} 0 & \text{if } t = 1 \text{ (first timepoint)} \\ c_{bm} & \text{if on drug (raw BR} > 0) \\ \dfrac{\mu^{\mathrm{adj}}_{\mathrm{BR},t}}{\mu^{\mathrm{adj}}_{\mathrm{BR},t-1}} \cdot c_{bm} & \text{if off drug, raw BR} = 0,\ \mu^{\mathrm{adj}}_{\mathrm{BR},t} > 0 \\ 0 & \text{if off drug, } \mu^{\mathrm{adj}}_{\mathrm{BR},t} = 0 \end{cases}$$

In the R code:

```
if (p > 1) {
  correlations["bm", n1] <- correlations[n1, "bm"] <-
    ifelse(brtest[p],
           ifelse(brmeans[p] == 0, 0,
                  (mm1 / mm0) * modelparam$c.bm),
           modelparam$c.bm)
}
```

where `brtest[p]` is `TRUE` when the raw (pre-carryover) BR mean is zero at timepoint p.

4 The Artifact: Conditional Expectations Under the Original Rule

4.1 Conditional Mean of BR Given Biomarker

In a multivariate normal distribution, the conditional expectation of BR_t given a specific biomarker value bm is:

$$E[\mathrm{BR}_t \mid \mathrm{BM} = bm] = \mu_{\mathrm{BR},t} + \frac{\rho \cdot \sigma_{\mathrm{BR}}}{\sigma_{\mathrm{BM}}} \cdot (bm - \mu_{\mathrm{BM}}) \tag{1}$$

The key observation is that this conditional expectation depends on ρ and σ_{BR} but *not* on $\mu_{\mathrm{BR},t}$ (which only shifts the intercept). When $\rho = c_{bm} = 0.6$ and $\sigma_{\mathrm{BR}} = 8$, $\sigma_{\mathrm{BM}} = 15.36$:

$$E[\mathrm{BR}_t \mid \mathrm{BM} = bm] = \mu_{\mathrm{BR},t} + 0.3125 \cdot (bm - 124.3)$$

The slope 0.3125 is independent of $\mu_{\mathrm{BR},t}$.

4.2 Numerical Example: N-of-1 Path A at BD3

Consider timepoint BD3 in the N-of-1 design (Path A), where the participant has been off drug for 1 week with carryover half-life $t_{1/2} = 0.1$ weeks:

- Raw BR mean: $\mu^{\mathrm{raw}}_{\mathrm{BR},5} = 0$ (off drug, tod = 0)
- Adjusted BR mean: $\mu^{\mathrm{adj}}_{\mathrm{BR},5} = 0.01$ (negligible carryover residual: 10.19×2^{-10})
- BR standard deviation: $\sigma_{\mathrm{BR}} = 8$ (unchanged regardless of mean)

Under the original rule, $\rho = 0.6$ because $\mu^{\mathrm{adj}}_{\mathrm{BR},5} \neq 0$. Applying Equation 1:

Table 1: Conditional $E[\mathrm{BR}_5 \mid \mathrm{BM}]$ at BD3 under the original correlation rule

Biomarker value	BM deviation	E[BR] (original)	E[BR] (revised)
109 (−1 SD)	−15.4	−4.8	+0.01
124 (mean)	0	+0.01	+0.01
140 (+1 SD)	+15.7	+4.9	+0.01

Under the original rule, the conditional BR spans a range of nearly 10 points (−4.8 to +4.9) centered on a mean of 0.01. The biomarker predicts a large, clinically meaningful difference in drug effect at a timepoint where the drug effect has decayed to 0.01% of its peak.

Under the revised rule, $\rho \approx 10^{-6}$, and the conditional BR is 0.01 for all biomarker values.

4.3 The Source of the Artifact

The artifact arises from a disconnect between two elements of the multivariate normal specification:

1. **The mean vector** reflects the drug effect decaying to near zero via carryover.
2. **The correlation matrix** maintains full biomarker–BR correlation ($\rho = 0.6$) as long as the adjusted mean is nonzero.
3. **The standard deviation vector** is fixed at $\sigma_{\mathrm{BR}} = 8$ regardless of the mean.

The combination of full correlation and fixed variance around a near-zero mean produces draws where some participants (high biomarker) show substantial positive BR and others (low biomarker) show substantial negative BR, despite the mean drug effect being negligible. The negative BR values are particularly problematic: they imply that the drug is *harming* low-biomarker participants during the off-drug period, which has no clinical basis.

4.4 Why This Destroys Power

The LME analysis model for crossover designs is:

$$S_{i,t} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot D_{bc,t} + \beta_3 \cdot t + \beta_4 \cdot bm_i \cdot D_{bc,t} + u_i + \varepsilon_{i,t}$$

where D_{bc} is the drug indicator and β_4 is the biomarker–treatment interaction of interest.

The spurious BM–BR correlation during off-drug periods introduces structured noise into the $bm \cdot D_{bc}$ term. When the participant is off drug ($D_{bc} = 0$ or near zero), the model expects no biomarker–drug interaction. But the correlated BR draws create biomarker-specific variation that inflates the residual variance of the interaction estimate, increasing the standard error of $\hat{\beta}_4$ and reducing the probability of rejecting $H_0 : \beta_4 = 0$.

The effect is largest for designs with many off-drug timepoints (N-of-1 has 4 out of 8) and smallest for designs with no off-drug periods (OL is unaffected because all timepoints are on drug).

5 Clinical Evaluation

5.1 Is the Original Rule Clinically Reasonable?

The original rule implies the following clinical scenario:

> One week after discontinuing a drug whose pharmacologic effect has decayed to 0.01% of its peak value, a patient's baseline blood pressure still perfectly predicts whether the patient is "responding" (positive BR) or "not responding" (negative BR) to the drug.

This is not clinically plausible. The biomarker–treatment interaction is mediated through the drug's mechanism of action. When the drug effect has decayed to negligible levels, the biomarker's predictive value for the drug effect should also decay. The step-function behavior of the original rule (full predictive power until the last molecule clears, then instantly zero) does not correspond to any known pharmacologic mechanism.

Furthermore, the negative conditional BR values for low-biomarker participants are clinically incoherent. They imply that discontinuing the drug *worsens* symptoms beyond the natural trajectory, and that this worsening is specific to low-biomarker individuals. No such rebound effect was hypothesized in the original publication.

5.2 Is the Revised Rule Clinically Reasonable?

The revised rule implies:

> The biomarker's predictive power for the drug effect decays in exact proportion to the drug effect itself. When the drug effect is at 50% of peak, the biomarker–BR correlation is $0.5 \cdot c_{bm}$. When the drug effect is at 0.01%, the correlation is $0.0001 \cdot c_{bm}$.

This is more clinically reasonable but may be overly optimistic. It assumes perfect proportional tracking between the drug effect magnitude and the biomarker's predictive value. In practice, the relationship might be non-linear: the biomarker might retain some predictive value even as the drug effect wanes (e.g., if the biomarker indexes a slow-to-reverse physiological state that mediates the drug response).

However, the practical difference between the revised rule and any plausible intermediate model is negligible at the carryover half-lives tested ($t_{1/2} = 0.1$ and 0.2 weeks). At these values, the carryover residual at the next assessment timepoint (1–4 weeks later) is less than 0.1% of the peak BR, and any correlation proportional to the residual would be effectively zero.

5.3 Summary of Clinical Assessment

Table 2: Clinical plausibility of the two correlation rules

Criterion	Original	Revised
Drug effect near zero ⇒ BM predictive?	Yes (full)	No (≈ 0)
Negative BR during off-drug periods?	Yes (for low BM)	No
Proportional decay of predictive power?	No (step function)	Yes (exact)
Clinically plausible?	No	Approximately

6 Statistical Evaluation

6.1 Covariance Matrix Coherence

In a well-specified multivariate normal model, the correlation structure should be consistent with the mean structure. When $\mu_{\mathrm{BR},t} \approx 0$, applying a large correlation with BM creates a conditional distribution where most of the conditional variation is biomarker-driven rather than drug-driven. The conditional variance of BR given BM is:

$$\mathrm{Var}(\mathrm{BR}_t \mid \mathrm{BM}) = \sigma^2_{\mathrm{BR}}(1 - \rho^2) = 64 \cdot (1 - 0.36) = 40.96$$

The conditional standard deviation is $\sqrt{40.96} = 6.4$, which is enormous relative to the mean of 0.01. The coefficient of variation is 640, indicating a distribution centered at effectively zero but with substantial spread driven by the biomarker conditioning. This is statistically valid (the MVN can produce such draws) but represents a pathological parameterization where the correlation structure dominates the mean structure at off-drug timepoints.

6.2 Impact on the Covariance Matrix Positive Definiteness

The original rule can create situations where the correlation between BM and BR changes abruptly between adjacent timepoints (e.g., from 0.6 at BD2 to 0.6 at BD3 to 0.6 at BD4, despite BD3–BD4 being off drug). This does not violate positive definiteness per se, because the same correlation is applied to all off-drug timepoints with any residual. However, the step function creates discontinuities in the correlation structure that increase the likelihood of requiring the `make.positive.definite()` correction, which in turn perturbs the intended correlation structure.

6.3 Effect Size Bias

The spurious BM–BR correlation during off-drug periods does not bias the point estimate of β_4 (the interaction coefficient) in expectation, because the noise is symmetric around zero. However, it inflates the standard error of $\hat{\beta}_4$ by introducing structured residual variance that the random effects cannot absorb. The practical consequence is not bias but loss of precision, hence loss of power.

6.4 Type I Error

At $c_{bm} = 0$ (no true interaction), neither rule produces spurious correlations, because both rules multiply c_{bm} by some factor. When $c_{bm} = 0$, any multiplicative scaling yields zero. Consequently, the Type I error rate is approximately 0.05 under both rules, consistent with the publication's findings and the present simulation results.

7 Implications for Published Results

7.1 Which Results Are Affected?

Only results involving nonzero carryover ($t_{1/2} > 0$) and designs with off-drug timepoints are affected. Specifically:

- **Figure 4A** (carryover = 0): **Not affected.** When carryover is zero, there are no carryover residuals, and both rules assign $\rho = 0$ to off-drug timepoints with $\mu_{\text{BR}} = 0$.
- **Figure 4B** (carryover > 0): **Affected.** The power drop for N-of-1 and OL+BDC is driven by the correlation artifact, not by the carryover residual itself.
- **OL design**: **Not affected.** All timepoints are on drug, so no off-drug correlation assignment occurs.

7.2 Is the Paper's Central Conclusion Still Valid?

The paper's central conclusion is that N-of-1 designs are sensitive to carryover effects. Under the revised correlation rule, this conclusion does not hold: N-of-1 power is robust to carryover at the tested half-lives.

However, the paper's conclusion may still be qualitatively valid for a different reason. The carryover *mean* residual at $t_{1/2} = 0.1$ weeks is negligible (0.01%), but at longer half-lives (e.g., $t_{1/2} = 2$ weeks), the residual would be substantial:

$$\mu^{\text{adj}}_{\text{BR},t} = 10.19 \times (1/2)^{1/2} \approx 7.2 \quad \text{at } t_{1/2} = 2, t_{sd} = 1$$

At such half-lives, even the revised correlation rule would assign a substantial BM–BR correlation ($\sim 0.7 \times c_{bm}$), and the carryover would genuinely confound the analysis by maintaining a biomarker-specific drug effect during off-drug periods. The paper's finding would hold, but at longer carryover half-lives than those tested.

7.3 Recommendation

The published results at carryover half-lives of 0.1 and 0.2 weeks should be interpreted with caution. The dramatic power loss reported is primarily an artifact of the correlation assignment rule, not a genuine statistical consequence of carryover at those half-lives.

Future simulations should either:

1. Use the revised (proportional) correlation rule and test carryover at longer, clinically realistic half-lives (e.g., 1–4 weeks), or
2. Explicitly model the correlation decay function as a separate parameter, allowing sensitivity analysis over different assumptions about whether carryover effects retain biomarker specificity.

8 A Principled Correlation Decay Model

Neither the step function (original) nor exact proportional scaling (revised) is fully satisfactory as a general model. A more principled approach would parameterize the correlation decay explicitly:

$$\rho(\mathrm{BM}, \mathrm{BR}_t) = \begin{cases} c_{bm} & \text{if on drug} \\ c_{bm} \cdot g(t_{sd}; \boldsymbol{\theta}) & \text{if off drug} \end{cases}$$

where $g(t_{sd}; \boldsymbol{\theta})$ is a decay function with parameters $\boldsymbol{\theta}$ controlling the rate at which biomarker predictive power decays after drug discontinuation. Natural choices include:

- **Step function** (original): $g(t_{sd}) = \mathbf{1}[\mu^{\mathrm{adj}}_{\mathrm{BR},t} > 0]$
- **Proportional** (revised): $g(t_{sd}) = \mu^{\mathrm{adj}}_{\mathrm{BR},t} / \mu^{\mathrm{adj}}_{\mathrm{BR},t-1}$
- **Exponential decay**: $g(t_{sd}) = \exp(-\lambda \cdot t_{sd})$ with separate half-life λ for the correlation
- **Power law**: $g(t_{sd}) = (1 + t_{sd})^{-\alpha}$

The exponential decay model is particularly attractive because it allows the correlation half-life to differ from the mean effect half-life, which is pharmacologically plausible: the biomarker's predictive power might decay faster or slower than the drug effect depending on the mechanism of action.

This parameterization would allow sensitivity analysis over λ and separate the question of "how long does the drug effect last?" from "how long does the biomarker remain predictive?"

9 Conclusions

1. The power drop reported in Hendrickson et al. (2020) Figure 4B at carryover half-lives of 0.1 and 0.2 weeks is primarily driven by a modeling artifact in the correlation assignment rule, not by the carryover residual in the mean structure.
2. The artifact arises because the original code applies full biomarker–BR correlation ($\rho = c_{bm}$) to off-drug timepoints where the adjusted BR mean is negligible, creating clinically implausible conditional expectations (participants with low biomarkers show drug "harm" during off-drug periods).
3. The Ron Thomas proportional scaling (commit `8609f12`) corrects this artifact by scaling the correlation proportional to the carryover residual. This is clinically and statistically more defensible than the step-function rule.
4. The paper's qualitative conclusion (that N-of-1 designs are sensitive to carryover) may still hold at longer, more clinically realistic carryover half-lives, but the quantitative results at $t_{1/2} = 0.1$ and 0.2 weeks should be interpreted with caution.

5. Future implementations should parameterize the correlation decay function independently from the mean decay function, allowing explicit sensitivity analysis over assumptions about whether carryover drug effects retain biomarker specificity.

Appendix: Commit Chronology and Code Diffs

The following commits modified the core DGP files (`generateData.R` and `lme_analysis.R`):

`42ac030` (Initial commit) Published code. BM–BR correlation assigned as step function on adjusted BR mean. No scale factor on carryover decay.

`8609f12` (Ron Thomas edits) Three changes: (1) added scale factor $s = 2$ on carryover decay, (2) replaced step-function correlation rule with proportional scaling, (3) added timepoint-1 zero-correlation convention. Also added `full_model_out` option to `lme_analysis()`.

`f6ee86d` (Autocorrelation typo fix) Fixed a symmetry bug in the cross-factor correlation matrix (`correlations[n2,n1]` was incorrectly written as `correlations[n1,n2]`). No impact on Figure 4 because `make.positive.definite()` had been implicitly correcting the asymmetry.

`f70f86d` (Carryover term added) Added carryover functionality to other parts of the codebase. No changes to the core `generateData()` or `lme_analysis()` functions that affect Figure 4.

`325314f` (Modified Db to include carryover) Introduced continuous drug indicator `Dbc` in the analysis model. Contains two bugs preventing execution (typo in `op*carryover_scalefactor` and incorrect coefficient name `bm:DbcTRUE`).

`HEAD` (Current code) Fixed version of `325314f`. Uses continuous `Dbc` with exponential decay in the analysis model. Combined with the DGP changes from `8609f12`, produces the lowest correlation with the published Figure 4 ($r = 0.613$).

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